

Euclidean distance

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1 Rules

In this assignment you will be creating a program to measure Euclidean distance. You may know of this process (though not by this name) through use of the Pythagorean theorem. For this assignment you will need to devise a way to store a Cartesian coordinate; a function to perform the Pythagorean theorem; and finally a way to link these actions together. You may not use any libraries for this assignment. There must be a write up turned in with this assignment explaining: your thinking, difficulties you had, and solutions to those challenges.

2 Difficulties

For this assignment I'm trying something new. There will be different difficulties to this assignment that will gain you different points.

2.1 Difficulty one

This is a simple Pythagorean theorem function. You only need make a function that executes the Pythagorean theorem on two Cartesian points. This will net you approx 75% score.

2.2 Difficulty two

The full completion of this assignment on the integers. This will net you approx 100% score.

2.3 Difficulty three

The full completion of this assignment on the real numbers. This will net you approx 120% score.

2.4 Difficulty four

The full completion of this assignment on the complex numbers. This will net you approx 150% score.

3 Turn-in

You will turn in a python file, for example *euclid.py*. This file will contain the python logic you have constructed to solve the problem above (to some level). The second item of your turn in is an English write-up of your thoughts, challenges, and solutions during the assignment. Note that the English portion should be in the form of a **.pdf** file.